Joyal's Proof of Cayley's Formula

Deniz Göksu

November 17, 2026

Abstract

Cayley's formula is a result in enumerative combinatorics that gives the number of possible trees on a finite set of at least two labelled vertices. In this talk, we discuss André Joyal's proof of this result, which is a bijective proof that involves his notion of combinatorial species. We use Cayley's formula as a context to explore the basics of this notion, such as the arithmetic of species, the relation of species to generating functions, and how combinatorial interpretation benefits from the discipline of category theory.

MSC Number: 05A15, 05A19, 13F25, 18A25

Keywords: Cayley's formula, combinatorial species, Joyal, generating functions, formal power series, labelled trees, bijective proof, categorification

Address: *Department of Mathematics, Izmir Institute of Technology, Urla, Izmir, Turkey, dgoksu512@gmail.com*

References

[1] Cayley, A. (1889). A theorem on trees. *Quarterly Journal of Pure and Applied Mathematics, 23*, 376–378.

[2] Joyal, A. (1981). Uné théorie combinatoire des séries formelles. *Advances in Mathematics, 42*(1), 1–82. https://doi.org/10.1016/0001-8708(81)90052-9